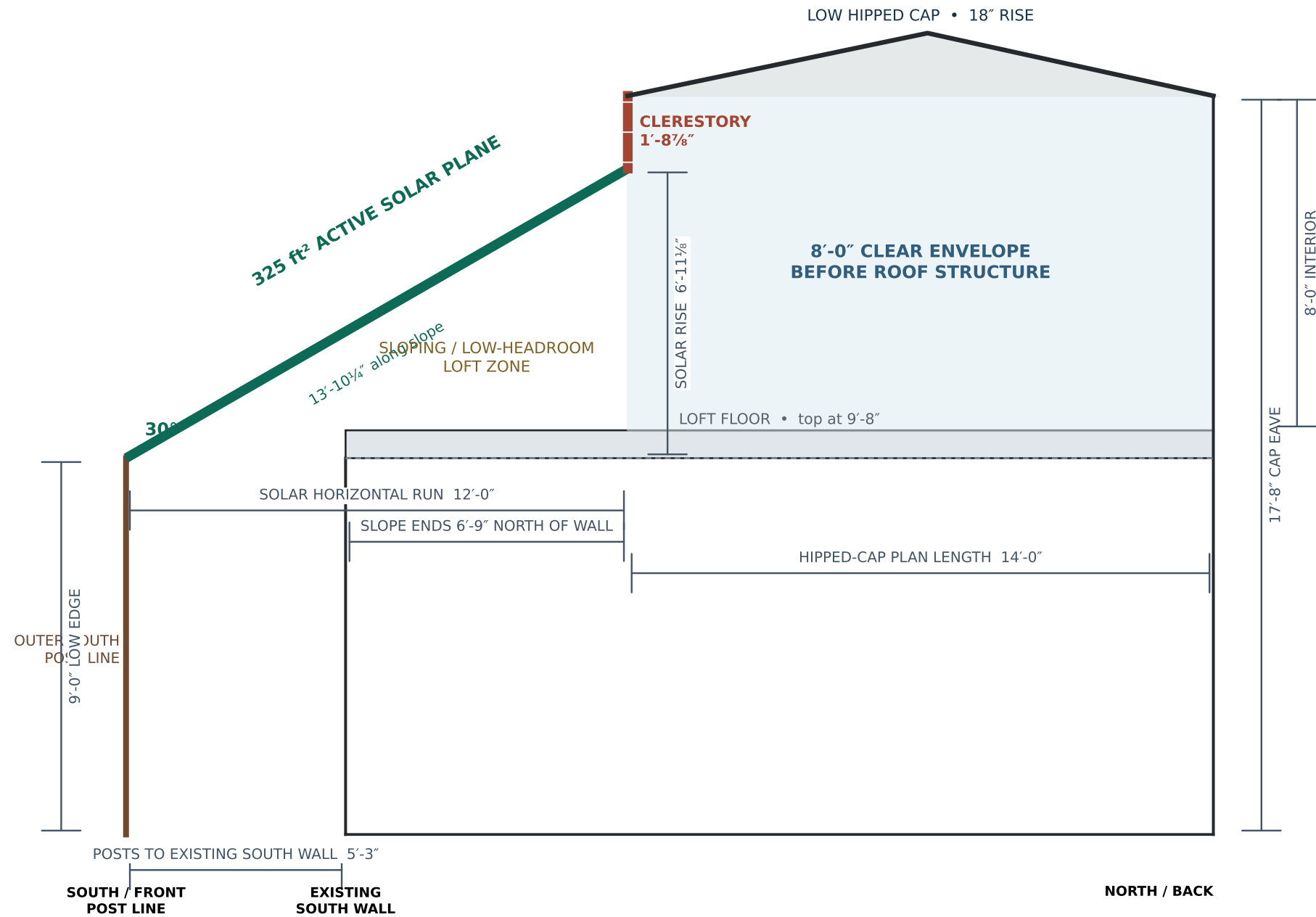


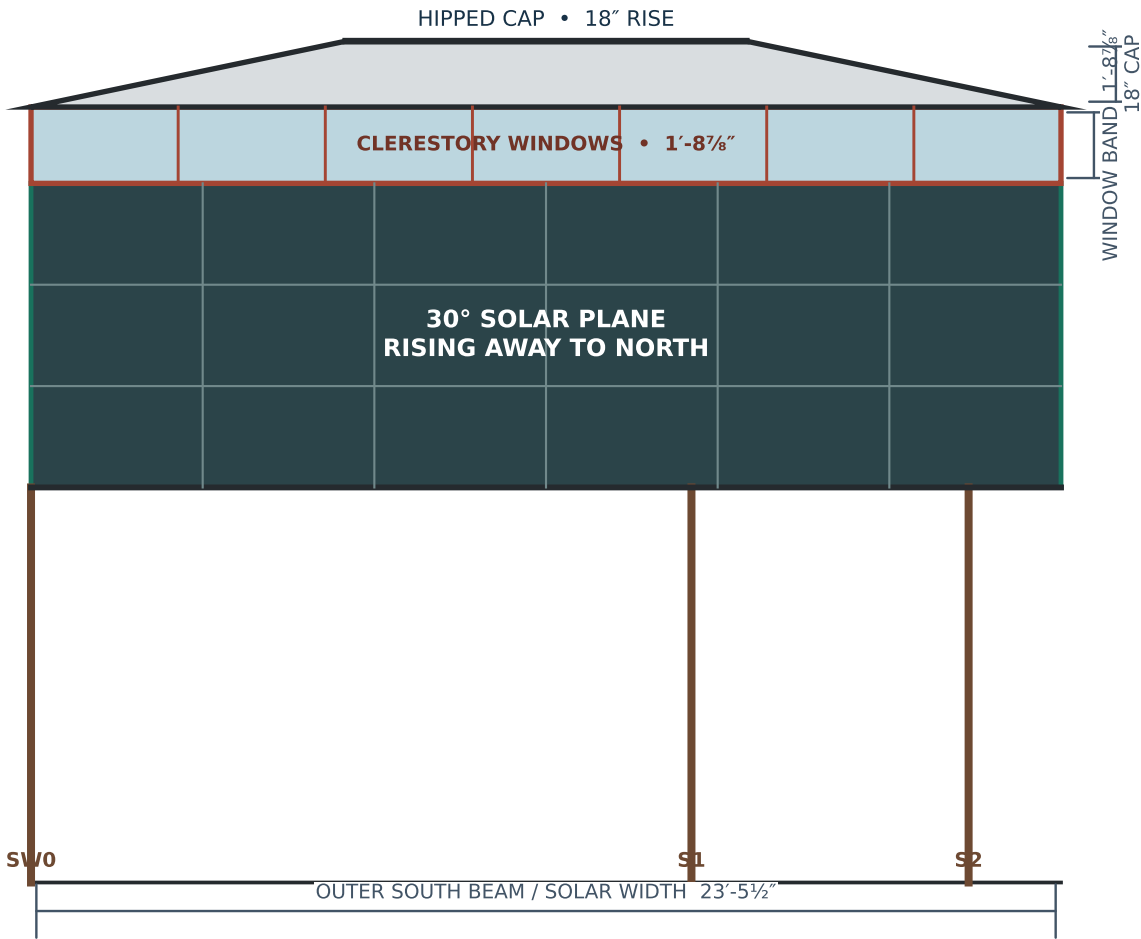
SOUTH-POST SOLAR SLOPE + CLERESTORY + LOW HIP CAP

30° solar plane • 325 ft² central study • 9 ft first-floor ceiling • 8 in floor assembly • 8 ft northern loft envelope

WEST ELEVATION / ENVELOPE SECTION



SOUTH ELEVATION



SOLAR-AREA CHECK

Solar area	Slope length	Plan run	Rise	Slope top	Pop to 8'
320 ft²	13'-7 ³ / ₄ "	11'-9 ³ / ₄ "	6'-9 ⁷ / ₈ "	15'-9 ⁷ / ₈ "	1'-10 ³ / ₈ "
325 ft²	13'-10 ¹ / ₄ "	12'-0"	6'-11 ¹ / ₈ "	15'-11 ¹ / ₈ "	1'-8 ⁷ / ₈ "
330 ft²	14'- ³ / ₄ "	12'-2 ¹ / ₄ "	7'- ³ / ₈ "	16'- ³ / ₈ "	1'-7 ⁵ / ₈ "

The 325 ft² version is the cleanest geometric fit:

- about 12 ft of horizontal run
- about 6 ft 11 in of rise
- about 20⁷/₈ in of clerestory to reach an 8 ft loft envelope

The solar roof begins at the T-SO / south-post line, 5 ft 3 in outside the existing south wall. The hipped cap is shown with an 18 in rise. Roof build-up, structural depths and drainage allowance will reduce clear height unless the cap datum is raised accordingly.